

Nernst Equation

An electrode potential is a measure of the extent to which the concentrations of the species in a half-cell differ from their equilibrium values. For example $\text{Ag}^+ + e^- \rightleftharpoons \text{Ag(s)}$ there is greater tendency for this process to occur in conc. solution of silver than in dilute solution of that ion. Nernst establishes quantitative relationship between potential and concentration. This potential dependence is describe by the Nernst equation

$$a_{\text{Ox}} + n e^- \rightleftharpoons b_{\text{Red}}$$

$$E = E^\circ - \frac{2.3026 RT}{nF} \ln \frac{[\text{Red}]^b}{[\text{Ox}]^a}$$

$$E = E^\circ - \frac{2.3026 RT}{nF} \log \frac{[\text{Red}]^b}{[\text{Ox}]^a}$$

Where

E° = the standard electrode potential at specific conc.

n = no. of electron involved in the half-reaction

R = Gas constant $8.314 \text{ J K}^{-1} \text{ mol}^{-1}$

T = absolute temperature

F = Faraday constant ~~0.05946~~ $96485 \text{ C per mole of electron}$

The value $2.3026 RT/F = 0.05916 \approx 0.0592$

Then Nernst equation $E = E^\circ - \frac{0.0592}{n} \log \frac{[\text{Red}]^b}{[\text{Ox}]^a}$

In terms of activity $E = E^\circ - \frac{0.0592}{n} \log \frac{a_{\text{Red}}}{a_{\text{Ox}}}$

Relationship between Activity and Concentration

The effective concentration of an ion in the presence of an electrolyte is called activity of the ion. The concentration of an ion is decreased by shielding if with other inert ions, and it represents the deficiency of the ion. The activity of an ion a_i is defined by

$$a_i = C_i f_i \quad \text{where}$$

C_i = Concentration of the ion

f_i = is the activity coefficient

The concentration is usually expressed as molarity, and the activity has the same units as the concentration.

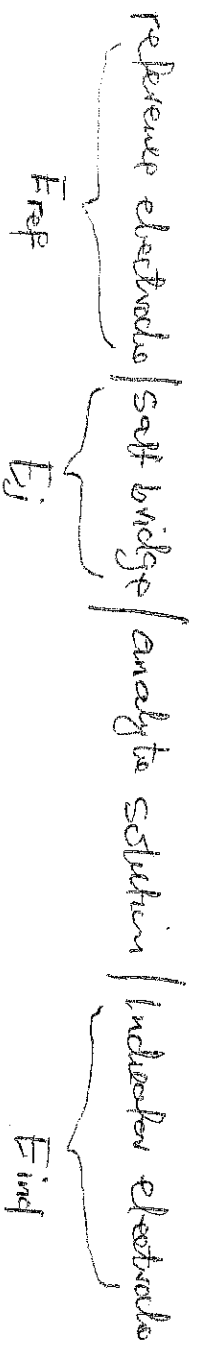
The activity coefficient is dimensionless, but numerical values for activity coefficient depend on the choice of standard state. The activity coefficient varies with the total no. of ions in solution and with their charge and it is a correction for interionic attraction.

In dilute solution, less than $10^{-4} M$, the activity coefficient of a simple electrolyte is near unity, and the activity is approximately equal to the concentration.

As the conc. of an electrolyte increases, or salt is added, the activity coefficient generally decreases, and the activity becomes less than the concentration. Activities are important in potentiometric measurement.

Cell for Potentiometric Measurements

In potentiometric measurements the activity of the ion^{is} to be calculated from the potential of the indicating electrode, the potential of the reference electrode will have₁ b_1^c known or determined. The cell can be represented as



The reference electrode is a half-cell with accurately known electrode potential E_{ref} , that is independent of the conc. of the analyte or any other ion in the solution under study. It is always treated as the left-hand electrode in potentiometric measurements. The indicator electrode, which is immersed in a solution of the analyte, develops a potential E_{ind} , that depends on the activity of the analyte. The third component of a potentiometric cell is the salt bridge that prevents the compartment of analyte solution from mixing with those of reference electrode. A potential is developed across the liquid junctions at each end of the salt bridge. These two potentials tend to cancel one another if the mobilities of the cation and anion in the bridge solution are approximately the same. The potential of the cell is given by the equation

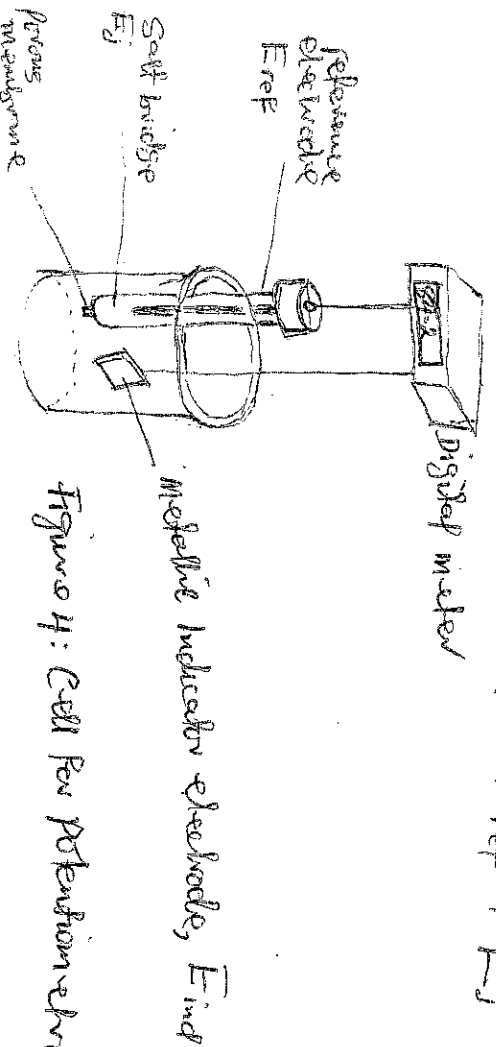
$$E_{\text{cell}} = E_{\text{ind}} - E_{\text{ref}} + E_j$$


Figure 4: Cell for potentiometric determinations.

Useful Potentiometry in Titration

A Potentiometric titration involves measurement of the Potential of a suitable indicator electrode as a function of titrant volume. Potentiometric titrations provide data that are more reliable than color. From titration that use chemical indicators and they are particularly useful with colored or turbid solution and for detecting the presence of unsuspended species.

(i) Complex-Formation Titration: - Both metallic and membrane electrodes have been used to detect end points in potentiometric titration involving complex formation.

(ii) Neutralization Titration: - Potentiometric neutralization titration are quite useful for analysing for analysing mixtures of acids or polyprotic acid. An approximate numerical value for the dissociation constant of a weak acid or base can be estimated from potentiometric titration curves.

(iii) Oxidation/Reduction Titration: - An inert indicator electrode constructed of platinum is used to detect end point in oxidation/reduction titration.

(iv) Potentiometric Determination of Equilibrium Constants: - Numerical values for solubility-product constant, dissociation constant, and formation constant are conveniently evaluated through the measurement of cell potentials.

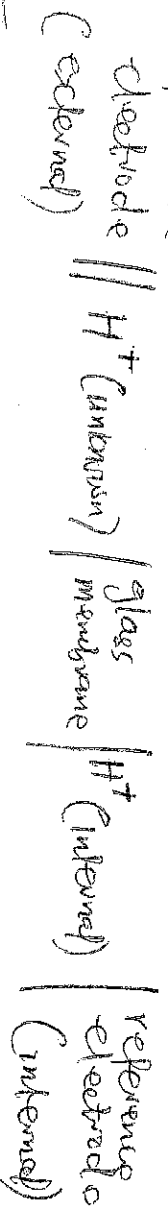
Ion-Selective Electrodes

Various types of membrane electrodes have been developed in which the membrane potential is selective toward a given ion or ions, just as the potential of the glass membrane of the glass electrode is selective toward hydrogen ion. Most of these electrodes is specific for a given ion, but each will possess

A certain selectivity toward a given ion or ions. Glass Membrane Ion Selective electrodes are suited to measurements in biological media. Ion selective electrodes can be made to be sensitive to three species Ca^{2+} , K^{+} , Cl^{-} , F^{-} , NO_3^{-} and H^{+} , ClO_4^{-} etc.

Types of Ion Selective Electrodes

(1) Glass Membrane (pH) :- It consists of a thin, H^{+} -sensitive glass membrane sealed into one end of a heavy-walled glass or plastic tube. A small volume of dilute HCl saturated with silver chloride ~~reference electrode~~ is contained in the tube. A silver wire in this solution forms a silver/silver chloride reference electrode which is connected to one of the terminals of a potential measuring device. If potential is constant and is set by the conc. of HCl . A complete cell is represented by



The potential of the glass is given by $E_{\text{glass}} = \text{constant} - \frac{2.303RT}{F} \log \frac{a_{\text{H}^{+}}^{\text{int}}}{a_{\text{H}^{+}}^{\text{ext}}}$
 The voltage of the cell is given by $E_{\text{cell}} = K + \frac{2.303RT}{F} \log a_{\text{H}^{+}}^{\text{ext}}$

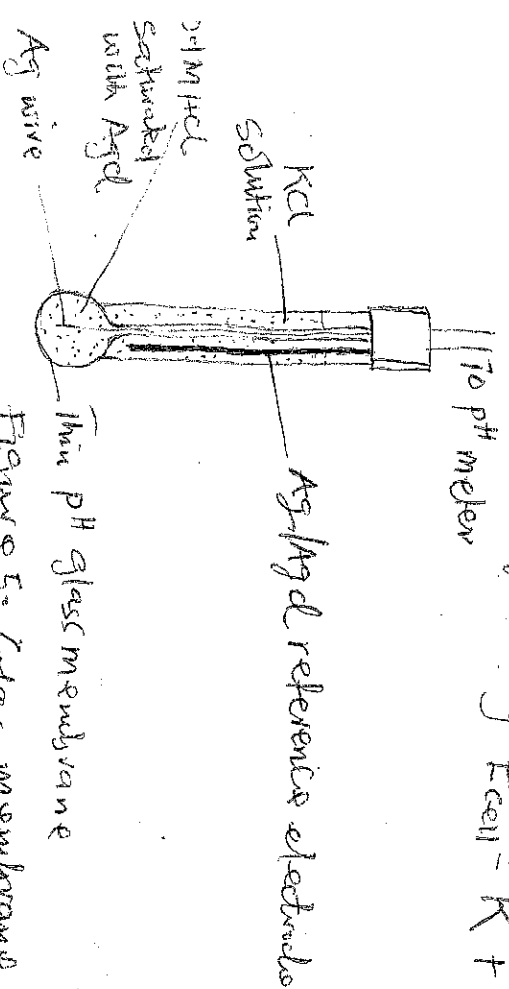
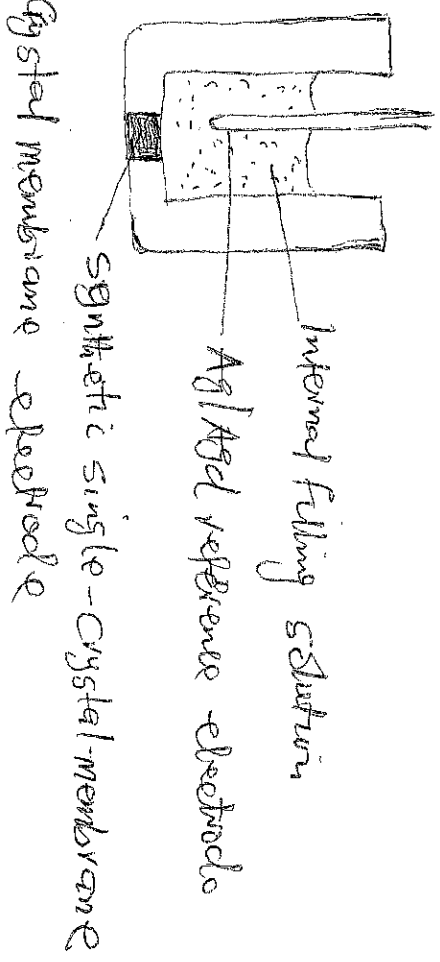


Figure 5- Glass membrane Electrode

(ii) Solid Membrane Electrode:-

The most successful example is the Fluoride electrode. The Membrane of a single crystal of lanthanum fluoride doped with some europium (11) to increase the conductivity of the crystal. ~~This electrode~~ Lanthanum fluoride is very stable, and this electrode exhibits Nernstian response to fluoride down to $10^{-5}M$ and non-Nernstian response down to $10^{-6}M$ ($19 ppb$). This electrode has at least a 1000-fold selectivity for fluoride ion over chloride, bromide, iodide, nitrate, sulphate, monohydrogen phosphate and bicarbonate ions and a 10-fold selectivity over hydroxide ion. The pH range is limited by the formation of hydrofluoric acid at the acid end and by hydroxide ion response at the alkaline end, a pH range of 4 to 9 can be measure.



Another Solid membrane electrode is based on Ag_2S membrane which respond to either Ag^+ or S^{2-} . Other Ag_2S mixed crystal electrodes are available for Cl^- , Br^- , SCN^- , Cu^+ , Pb^{2+} and cd^{2+} .

(ii) Liquid Membrane Electrode:

The potential of liquid-membrane electrode develops across the interface between the solution containing the analyte and a liquid ion exchanger that selectively binds with the analyte ion. Example of liquid membrane ion selective electrode is the Calcium liquid membrane electrode. It consists of a conducting membrane that selectively binds Calcium ions, an internal solution containing ~~membrane~~ that a fixed conc. of Calcium chloride, and a silver electrode that coated with AgCl to form an internal reference electrode. A potential develops across the membrane when the extent of the ion-exchanger dissociation at one surface differs from that of the internal and external solutions. The relationship between the membrane potential and the Calcium ion activity is given

$$\log E_b = E_1 - E_2 = \frac{0.0592}{2} \log \frac{a_1}{a_2}$$

Where a_1 = activity of Calcium ion in external analyte

a_2 = activity of Calcium ion in internal standard solution
Liquid membrane electrode for other ions like K^+ , Na^+ and BF_4^- are available.

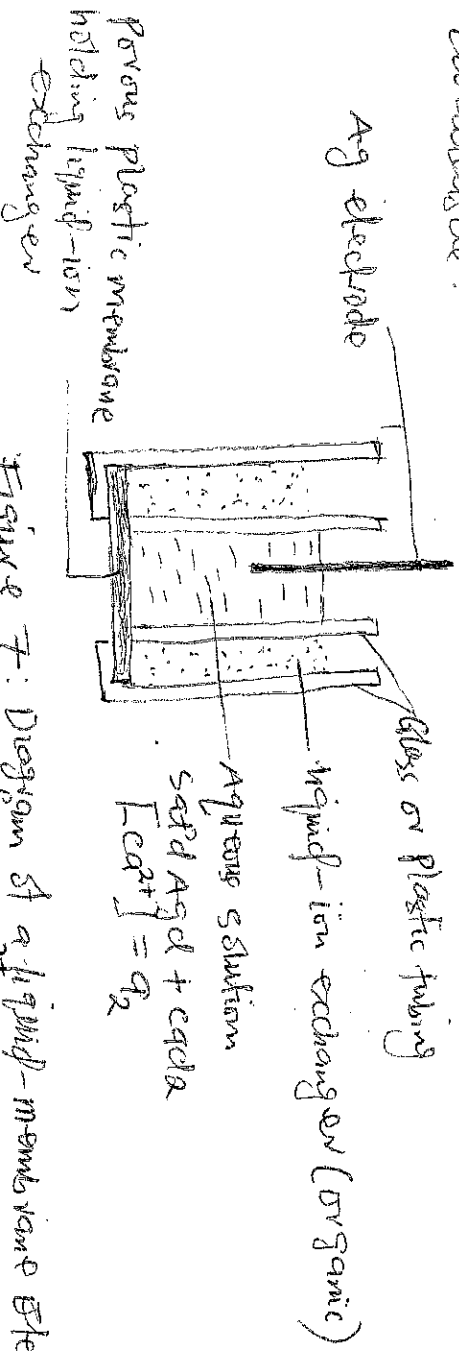


Figure 7: Diagram of a liquid-membrane electrode

To get ionic strength adjustment buffer (TISAB) is used to control the ionic strength and the pH of samples and standards in ion-selective electrode measurements.

VOLTAMMETRY:

Voltammetric method are based on measurement of current as a function of the potential applied to a small electrode. Mers electrode is an electrode with areas on the micrometer scale. Voltammetry is made up of three electrodes. Supporting electrode:- is a salt added in excess to the analyte solution to reduce the effect of migration and lowers the resistance of the solution.

Working electrode:- is the electrode at which the analyte is oxidized or reduced.
Counter electrode:- Electrolysis current passes between the working electrode and a counter electrode.

Principle of Voltammetry

The voltammetric cell is made up of three electrodes immersed in a solution containing the analyte and an excess of a non reactive electrolyte called a supporting electrolyte. One of the three electrodes is the working electrode, whose potential versus a reference electrode is varied linearly with time. The dimension of working electrode are kept small to enhance its tendency to become polarized. The reference electrode has a potential that remains constant. The third electrode is the counter electrode, which is often a coil of platinum wire or a pool of mercury.

The current in the cell passes between the working electrode and the counter electrode. A voltammogram is recorded by moving the contact (C) and recording the resulting current as a function of the potential E in the working electrode and the reference electrode. In modern voltammetric instruments, however, the excitation signal are generated electronically. These instruments vary the potential in a systematic way with respect to the ref. electrode and record the resulting current.

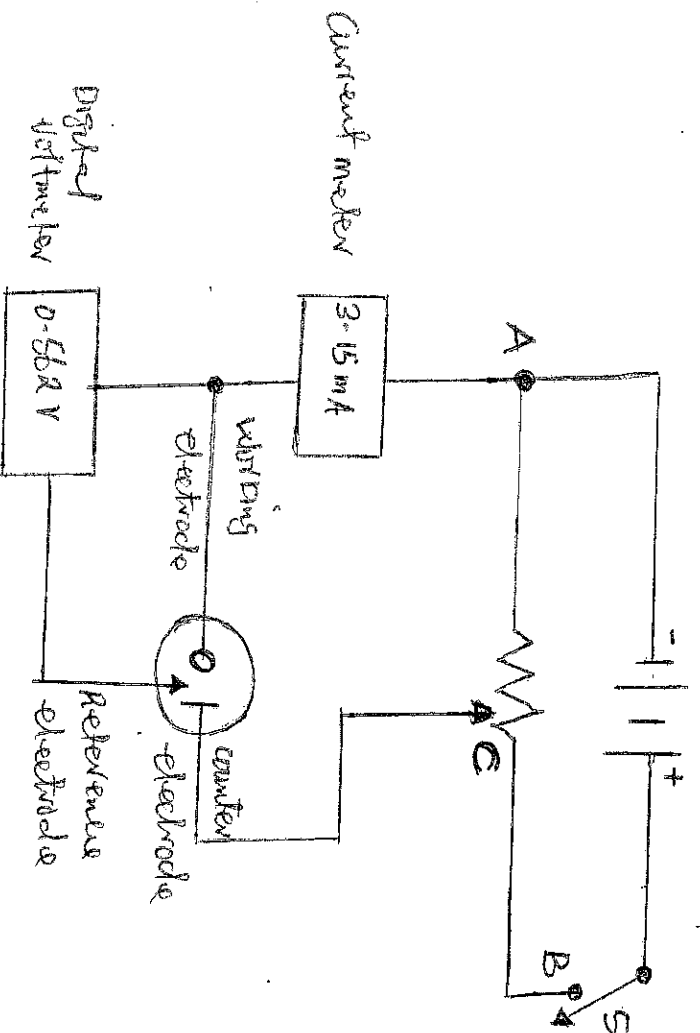
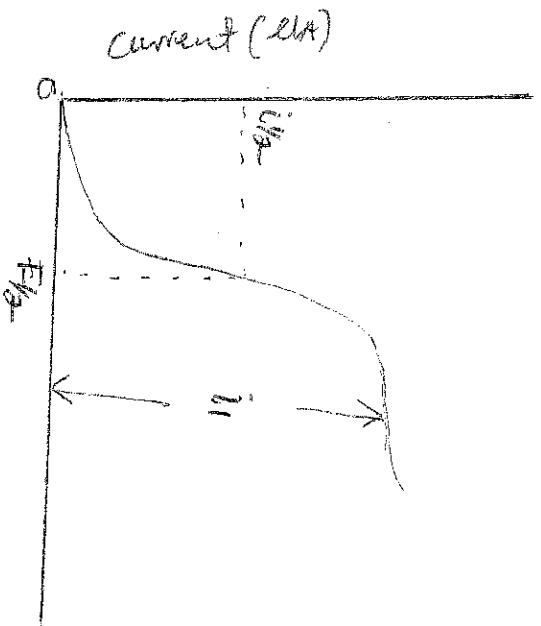


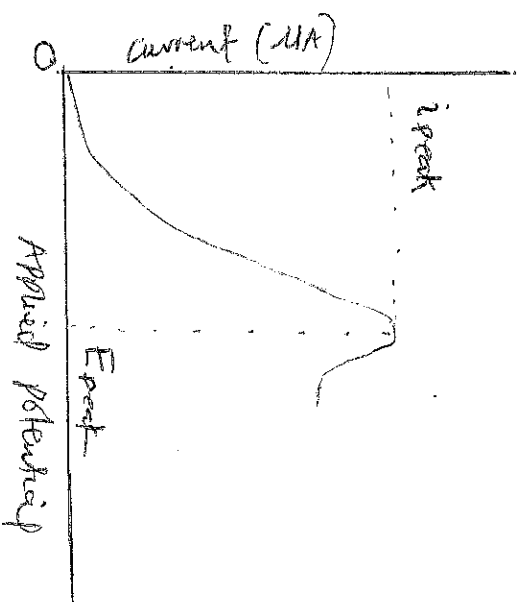
Figure 8: A Schematic system for manual potential scan voltammetry.

Shapes of Voltammetric Curves

Linear - Sweep Voltammograms under slow sweep condition generally have the shape of a sigmoidal (S-shaped) curve called a voltammetric wave. A voltammetric wave is an S-shaped wave obtained in current-voltage plots in voltammetry.



(a) Voltammetric curve of stirred solution or rotated electrode



(b) unstirred solution

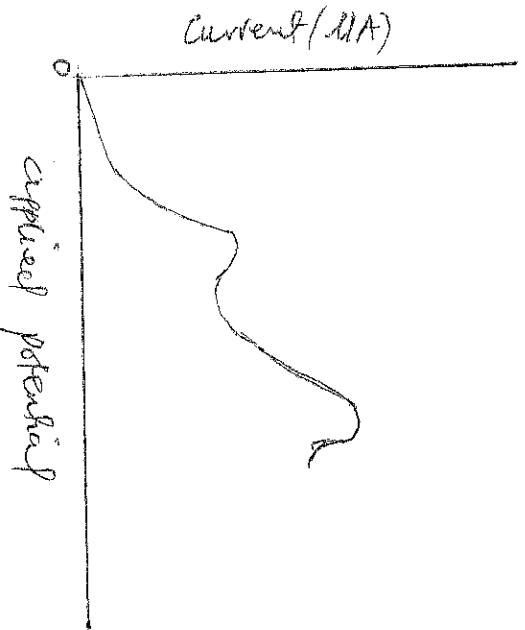


Figure 9: Voltammetric curves

(c) Stepwise reduction / oxidation of analyte

~~Diffusion~~ ^{limiting} current (i_L) - This is limited by the rate at which the reactant can be brought to the electrode surface by mass transport processes. Limiting current are generally directly proportional to reactant conc. Thus, we may write $i_L = K C_A$ where C_A = the analyte conc. and K = a constant

Quantitative linear sweep voltammetry is based on this relationship. Limiting current in voltammetry is the current plateau that is observed at the top of the voltammetric wave.